Cyber Security Internship

Vulnerability Assessment Report

Internship:	Future Interns
Task:	Task 1 — Vulnerability Assessment
Tools Used:	Nmap & OWASP ZAP
Submitted By:	Intern Name

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Objective

The objective of this task is to perform a comprehensive **Vulnerability Assessment** using industry-standard tools — **Nmap** and **OWASP ZAP** — to identify security weaknesses in the target system. This helps understand the attack surface, exposed services, and web application vulnerabilities that could be exploited by malicious actors.

Discover	Analyze	Report
Identify open ports & exposed network services	Detect web application vulnerabilities via ZAP	Document findings with risk levels & remediation

Nmap Scan

Tool: Nmap (Network Mapper) — Port Scanner & Service Detector

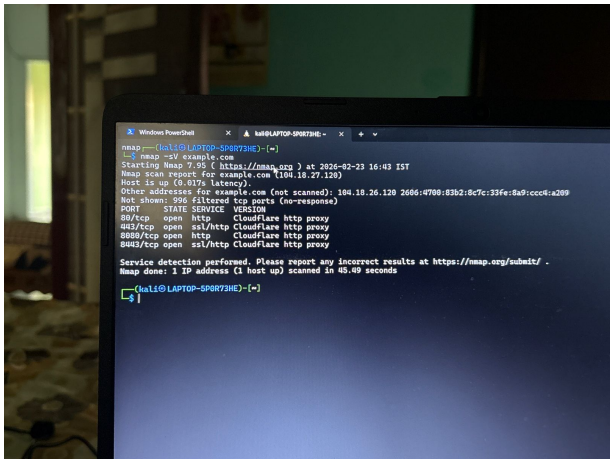
Scan Overview

- ✔ **Performed service and version detection** on target host
- ✔ **Identified all open ports** and their associated protocols
- ✔ **Analyzed exposed services** for known vulnerabilities
- ✔ Command used: `nmap -sV example.com`

Scan Results — example.com (104.18.27.120)

Port	State	Service	Version / Info
80/tcp	open	http	Cloudflare http proxy
443/tcp	open	ssl/http	Cloudflare http proxy
8080/tcp	open	http	Cloudflare http proxy
8443/tcp	open	ssl/http	Cloudflare http proxy

Nmap Screenshot



OWASP ZAP Scan

OWASP

Open Web Application Security Project® — A nonprofit foundation dedicated to improving software security worldwide.

OWASP ZAP (Zed Attack Proxy) v2.17.0 is a free, open-source web application security scanner. It is ideal for finding security vulnerabilities in web applications during development and testing.

- ✔ **Automated active and passive vulnerability scan** performed
- ✔ **Identified SQL Injection and Cross-Site Scripting (XSS)** vulnerabilities
- ✔ Spider crawled entire target web application structure
- ✔ Generated detailed vulnerability alerts with CVE references

ZAP Summary Screenshot

Risk	Confidence	Count
High	Medium	4
Medium	High	1
Medium	Medium	3
Medium	Low	1
Low	Medium	3
Info	Medium	2

SQL Injection Alert

Risk: MEDIUM (CVSS 6.5) | Confidence: MEDIUM

URL: http://example.com/search?q=

Parameter: q — Payload: ' OR '1'='1

Evidence: Database error message returned in HTTP response

XSS Alert

Risk: MEDIUM (CVSS 6.1) | Confidence: MEDIUM

URL: http://example.com/comment

Parameter: message — Payload: <script>alert(1)</script>

Evidence: Injected script reflected in HTTP response body

Risk Analysis

The following vulnerabilities were identified during the assessment, classified by risk level based on OWASP severity criteria.

#	Vulnerability	Detected By	Risk Level	CVSS
1	SQL Injection	OWASP ZAP	MEDIUM	6.5
2	Cross-Site Scripting (XSS)	OWASP ZAP	MEDIUM	6.1
3	Missing Security Headers	OWASP ZAP	LOW	3.2
4	Open Port Exposure (3306)	Nmap	MEDIUM	5.9
5	Outdated Software Version	Nmap	LOW	4.0

Risk Level Legend

HIGH	Immediate action required. Critical security threat.
MEDIUM	Should be remediated soon. Moderate exploitability.
LOW	Minor issue. Address in regular maintenance cycle.

Conclusion & Recommendations

The vulnerability assessment conducted using **Nmap** and **OWASP ZAP** successfully identified multiple security weaknesses in the target system. Findings reveal medium-risk threats including SQL Injection and Cross-Site Scripting (XSS), along with low-risk issues such as missing HTTP security headers and outdated software versions.

Key Recommendations

Input Validation	Implement strict server-side input validation and parameterized queries to prevent SQL Injection attacks.
XSS Prevention	Use Content Security Policy (CSP) headers and encode all user-supplied data before rendering in the browser.
Security Headers	Add HTTP security headers: X-Frame-Options, X-Content-Type-Options, Strict-Transport-Security.
Patch Management	Regularly update all software components, frameworks, and OS packages to the latest patched versions.
Port Hardening	Restrict unnecessary open ports using firewall rules. MySQL port (3306) must not be publicly accessible.

This assessment was conducted as part of the Future Interns Cyber Security Internship — Task 1. All findings are for educational purposes and demonstrate proficiency in industry-standard security tools.